

Fortran Project - Week 1

Variables and Arithmetic - Measurement Calculator

Today's goal

Use variables, READ/PRINT, and arithmetic expressions to build the first version of the project.

1. Guided Coding

Study the example code below with your team. Type it into your editor and run it. Do not copy and paste.

```
program week1_guided
  implicit none

  double precision :: x1, x2
  double precision :: average

  print *, 'Enter measurement 1:'
  read *, x1

  print *, 'Enter measurement 2:'
  read *, x2

  average = (x1 + x2) / 2.0d0

  print *, 'Average = ', average

end program week1_guided
```

2. Team Coding Tasks

1. Add a third measurement x3.
2. Modify the average so that it uses all three measurements.
3. Print the three input values and the final average.
4. Compile and run the program with 10, 20, 30. Confirm that the average is 20.

3. Expected Output / Checkpoint

```
Enter measurement 1:
10
Enter measurement 2:
20
Enter measurement 3:
30
Average = 20.0
```

4. Submission Check

- The program compiles without errors.
- The program runs from beginning to end.
- Every team member can explain the code added this week.
- Save the completed source code as week1_project.f90.